MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 5RP2A AND 5RP7A 1/

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Type phosphor	Ef	Ec1	ed	Eb1	Eb2	Eb3	Rg	Zd	Eb3/Eb2	Alt
Unit:		V	Vdc	v	Vdc	Vdc	Vdc	Meg	Meg	Ratio	ft
Maximum:	P7A	6.9	0	1,200	1,700	3,850	28,850	1.5	1.0	10	35,000
	P2A	6.9	0	1,200	1,700	3,850	25,850	1.5	1.0	10	35,000
Minimum:	P2A, P7A	5.7	-125	---	---	1,000	5,000	---	---	---	---
TEST CONDITIONS: (See note 1)	P2A, P7A	6.3	Adj	---	Focus	2,000	4,000	---	---	---	---

GENERAL:

Qualification - Required

1/ See note 5

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1002	Barometric pressure, reduced	6	178 mmHg	---	---	---	---
5101	Neck and bulb alinement (electrostatic types)	-		Diam	---	2.25	inches
5216	Cathode illumination	-		---	---	---	---
(B) 1216	Base material insulating quality	-	Zone 5 (minimum)	---	---	---	---
5246	Focusing voltage (zero-bias)	-		Eb1	362	695	Vdc
5248	Deflection factor (1D2)	-	Eb3 = Eb2 = 2,000 Vdc	DF	60	90	Vdc/in.
5248	Deflection factor (3D4)	-	Eb3 = Eb2 = 2,000 Vdc	DF	60	90	Vdc/in.
5248	Deflection-factor uniformity	-		---	---	---	---
1331	Direct-interelectrode capacitance	-	Cathode to all Control grid to all D1 to D2 D3 to D4 D1 to all except D2 D2 to all except D1 D3 to all except D4 D4 to all except D3	Ck Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	--- --- --- --- --- --- --- ---	7.5 6.5 2.5 2.5 3.5 3.5 3.0 3.0	pF pF pF pF pF pF pF pF
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	1.0	mm
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas "cross"	7	Ib3 = 50 μAdc	---	---	---	---
5103	Pattern distortion	2, 3	Eb2 = 2,000 Vdc; Eb3 = 20,000 Vdc; Ib3 = 5 μAdc	Useful diam	3.0	---	inches
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
---	Persistance (Type 5RP2A)	4		---	---	---	---
(B) 5223	Modulation	-	Ib3 = 50 μAdc	ΔEc	---	45	Vdc
5231	Spot position (electrostatic deflection)	-		---	---	25	mm
5231	Spot displacement (leakage)	-		Displ	---	10	mm

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 1 -Continued						
5241	Grid cutoff voltage	-		Eco	-30	-90	Vdc
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Anode No. 2 leakage current	-		---	---	---	---
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
(B) 5201	Electrode current (anode No. 1)	-	Ec1 = 0	Ib1	-50	10	μAdc
5201	Electrode current (cathode)	-	Ib3 = 50 μAdc	Ik	---	1,000	μAdc
5101	Base alinement (electrostatic types)	-	ID2; pin No. 5	---	---	---	---
5101	Side terminal alinement (electrostatic types)	-	ID2	---	---	---	---
5101	Angle between traces	-		---	---	---	---
5101	Neck and base alinement (electrostatic types)	-		---	---	---	---
5216	Stray light emission (conventional types)	-	Eb2 = 3,850 Vdc; Eb3 = 2,585 Vdc	---	---	---	---
(B) 5221	Screens (Type 5RP7A)	-		---	---	---	---
5226	Line width A (electrostatic deflection)	-	Ib3 = 50 μAdc	Width	---	0.8	mm
5226	Line width B (electrostatic deflection)	-	Ib3 = 50 μAdc	Width	---	0.9	mm
5246	Focusing voltage at cutoff	-		Eb1	460	690	Vdc
5248	Deflection factor	-	1D2	DF	72	102	Vdc/in.
5248	Deflection factor	-	3D4	DF	71	107	Vdc/in.
1101	Secureness of base, cap, or insert	-		---	---	---	---
(B) 1111	Base pin solder depth	-		---	---	---	---
(B) 1105	Permanence of marking	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Ib3 = 5 μAdc; Eb2 = 3,850 Vdc; Eb3 = 25,850 Vdc	t	500	---	hrs
---	Life-test end points:						
5226	Line width A	-	Ib3 = 37.5 μAdc	Width	---	0.8	mm
5226	Line width B	-	Ib3 = 37.5 μAdc	Width	---	0.9	mm
5223	Modulation	-	Ib3 = 37.5 μAdc	ΔEc1	---	45	Vdc

NOTES:

1. All intensifier snap terminals connected together.
2. 25 megohm resistor connected between each intensifier and anode No. 2 snap terminal.
3. To be checked with a raster focused with the high frequency deflection on the D3D4 plates. Astigmatism as checked by a merging raster line width check shall not exceed a ratio of 3 to 1.
4. The persistence is specified as the Cb1 value as measured for P7 screens (Radiation Laboratory Report No. 62-7, pages 24, 25, dated 14 May 1943) at a Q of 20, corrected for an area of 50 sq cm to make the reading obtained comparable with the Cb1 values for P7 screens under the standard reference conditions. The Cb1 value shall not be less than 350 cb.
5. Tube types 5RP1A, 5RP4A, 5RP5A, and 5RP11A have been deleted. Tubes for replacement purposes shall be procured inaccordance with EIA data.
6. Maximum potentials applied to base, deflection plates, and A2 only.
7. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 17
DSA - ES

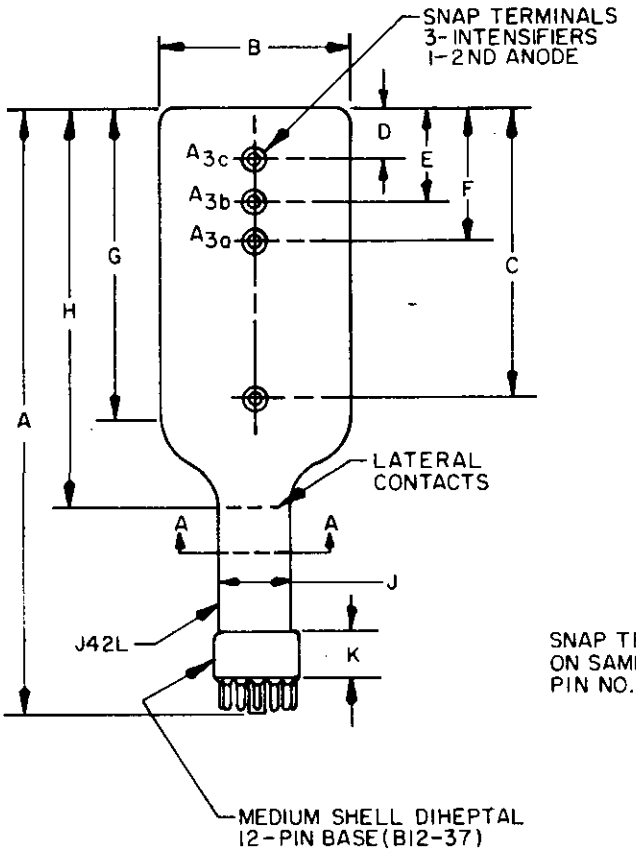
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force - 11

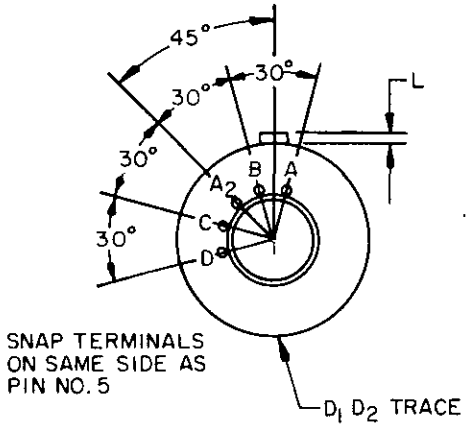
Preparing activity: Navy - EC

Agent: DSA - ES

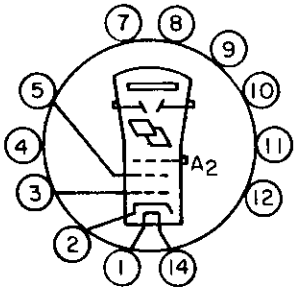
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SECTION A-A



PIN CONNECTIONS

Pin. No.	Element
1	Heater
2	Cathode
3	Control Electrode
4	Internal connection
5	Focusing Electrode
7	No connection
8	" "
9	" "
10	" "
11	" "
12	" "
14	Heater

Terminal A - Deflection plate D_4
B - " " D_3
C - " " D_1
D - " " D_2
A_2 - 2nd anode

FIGURE 1. Outline drawing of electron tube types 5RP2A and 5RP7A.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
Base: B12-37		
Quality conformance inspection, part 2		
A	16.375(415.93)	17.125(434.98)
B	5.156(130.96)	5.344(135.74)
C	7.750(196.85)	8.250(209.55)
D	1.188 (30.18)	1.688 (42.88)
E	2.312 (58.72)	2.812 (71.42)
F	3.438 (87.33)	3.938(100.03)
H	10.750(273.05)	11.250(285.75)
J	1.938 (49.23)	2.063 (52.40)
L		.125 (3.18)
Reference dimensions (see note e)		
G	8.500(215.90)	
K	1.312 (33.32)	

NOTES:
a. The angle between the trace produced by D_1D_2 and its intersection with the plane through the tube axis and pin No. 5 will not exceed 10°.
b. With D_1 positive with respect to D_2 the spot will be deflected toward pin No. 5, likewise with D_3 positive with respect to D_4 the spot will be deflected toward pin No. 2.
c. The angle between the trace produced by D_1D_2 and the trace produced by D_3D_4 will be 90° ± 3°.
d. The D_1D_2 deflecting plates are the pair nearer the screen.
e. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 5RP2A and 5RP7A-continued.

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